

Kanchan Chowdhury

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RESEARCH INTERESTS

Database Systems, Geospatial Data Analytics, and Machine Learning Systems

EDUCATION

- **Arizona State University** Tempe, Arizona
PhD & MS in Computer Science Aug. 2018 – July 2024
Advisor: Prof. Mohamed Sarwat & Prof. Jia Zou, CGPA: 4.00
- **Chittagong University of Engineering and Technology** Chittagong, Bangladesh
Bachelor of Science in Computer Science and Engineering Mar. 2010 – Nov. 2014
Advisor: Prof. Mohammed Moshikul Hoque, CGPA: 3.76

EXPERIENCE

- **Marquette University** Milwaukee, Wisconsin
Assistant Professor Aug. 2025 - Present
Department of Computer Science
 - **Responsibilities:**
 - 1) Conducting research in the intersection of AI, geospatial data, and databases.
 - 2) Mentoring graduate and undergraduate students.
 - 3) Teaching courses such as *Fundamentals of AI, Algorithms, and Advanced Algorithms*.
- **Arizona State University** Tempe, Arizona
Postdoctoral Research Scholar Aug. 2024 - Jun. 2025
Mentor: Prof. Jia Zou
 - **Responsibilities:**
 - 1) Conducting research on optimizing end-to-end pipelines consisting of database queries and machine learning models for inference workloads, reducing runtime and memory usage of in-database machine learning systems.
 - 2) Helping my mentor in writing research proposals for grants, reviewing research papers, and co-mentoring students.
- **Arizona State University** Tempe, Arizona
Research Assistant Aug. 2018 - July 2024
 - **Research Projects:**
 - 1) Co-optimization of machine learning and join queries based on model decomposition and join push-down.
 - 2) Designing and implementing a deep-learning and data processing system for raster imagery and vector datasets.
 - 3) Re-partitioning geospatial datasets to reduce spatial model training time and memory usage.
- **Wherobots Inc.** Scottsdale, Arizona
Research and Development Intern Jan. 2023 - Aug. 2023
 - **Responsibilities:** Designing and developing spatial machine learning and deep learning tools, scalable map-matching, and geospatial data analytical algorithms. Integrating the developed tools into the Wherobots cloud platform.
- **Gagagugu PTE LTD** Dhaka, Bangladesh
Software Engineer Jan. 2017 - Jun. 2018
 - **Responsibilities:** Developing Android Apps with social networking features such as calling, messaging, and posting.
- **Le Chef Plc** Dhaka, Bangladesh
Android Application Developer Jan. 2015 - Dec. 2016
 - **Responsibilities:** Developing Android Apps featuring online order and reservation services for restaurants in the UK.

SELECTED PUBLICATIONS

- Samya Acharja, **Kanchan Chowdhury**; Natural Language Interfaces for Spatial and Temporal Databases: A Comprehensive Overview of Methods, Taxonomy, and Future Directions. *IEEE Access*, 2026
- Lixi Zhou, **Kanchan Chowdhury**, Lulu Xie, Jaykumar Tandel, Hong Guan, Zhiwei Fan, Xinwei Fu, Jia Zou; CACTUSDB: Unlock Co-Optimization Opportunities for SQL and AI/ML Inferences. *Accepted to IEEE International Conference on Data Engineering (ICDE)*, 2026
- **Kanchan Chowdhury**, Lixi Zhou, Lulu Xie, Xinwei Fu, Jia Zou; INFERF: Declarative Factorization of AI/ML Inferences over Joins. *ACM SIGMOD International Conference on Management of Data*, 2026
- Hong Guan, Lei Yu, Lixi Zhou, Li Xiong, **Kanchan Chowdhury**, Lulu Xie, Xusheng Xiao, Jia Zou; Privacy and Accuracy-Aware AI/ML Model Deduplication. *ACM SIGMOD International Conference on Management of Data*, 2025
- **Kanchan Chowdhury**, Lulu Xie, Lixi Zhou, Jia Zou; ExBoost: Out-of-Box Co-Optimization of Machine Learning and Join Queries. *International Conference on Database Systems for Advanced Applications*, 2025
- **Kanchan Chowdhury**, Mohamed Sarwat; Deep Learning with Spatiotemporal Data: A Deep Dive into GeotorchAI. *40th International Conference on Data Engineering (ICDE)*, 2024
- Lixi Zhou, Qi Lin, **Kanchan Chowdhury**, Saif Masood, Jia Zou; Serving Deep Learning Models from Relational Databases. *27th International Conference on Extending Database Technology (EDBT)*, 2024
- **Kanchan Chowdhury**, Mohamed Sarwat; A Demonstration of GeoTorchAI: A Spatiotemporal Deep Learning Framework. *ACM SIGMOD International Conference on Management of Data*, 2023 (Demo Track)
- **Kanchan Chowdhury**, Vamsi Meduri, Mohamed Sarwat; A Machine Learning-Aware Data Re-partitioning Framework for Spatial Datasets. *38th International Conference on Data Engineering (ICDE)*, 2022
- **Kanchan Chowdhury**, Mohamed Sarwat; GeoTorch: A Spatiotemporal Deep Learning Framework. *30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*, 2022 (Short Paper)
- Vamsi Meduri, **Kanchan Chowdhury**, Mohamed Sarwat; Evaluation of Machine Learning Algorithms in Predicting the Next SQL Query From the Future. *ACM Transactions on Database Systems (TODS)*, 2021
- Jia Yu, **Kanchan Chowdhury**, Mohamed Sarwat; Tabula in Action: A Sampling Middleware for Interactive Geospatial Visualization dashboards. *Proceedings of the VLDB Endowment 13*, 2020 (Demo Track)
- Hong Guan, Ansh Tiwari, Summer Gautier, Rajan Hari Ambrish, Lixi Zhou, Deepti Gupta, Yancheng Wang, Yingzhen Yang, Chaowei Xiao, **Kanchan Chowdhury**, Jia Zou; Declarative Privacy-Preserving Inference Queries. *International Conference on Database Systems for Advanced Applications*, 2025 (Demo Track)
- Vamsi Meduri, **Kanchan Chowdhury**, Mohamed Sarwat; Recurrent Neural Networks for Dynamic User Intent Prediction in Human-Database Interaction. *International Conference on EDBT, 2019 (Short Paper)*

REVIEWING SERVICES

- **Conference Program Committee**: Served/serving in the program committee of following conferences: VLDB 2026, SIGSPATIAL 2026, ICDE 2027, and EDBT 2027.
- **Journal Paper Reviewer**: Served/serving as a reviewer for the following journals: ACM Transactions on Database Systems (TODS), IEEE Transactions on Knowledge and Data Engineering (TKDE), and Journal of Scientific Data.
- **Conference Paper Reviewer**: Served as a reviewer for the conferences ICCAD 2023 and DAC 2025.
- **External Reviewer**: Reviewed papers as an external reviewer for the following conferences and journals - SIGMOD 2020, 2022, 2025, PVLDB 2019-2025, ICDE 2020, SIGSPATIAL 2021, VLDB Journal, and TSAS Journal.

ADDITIONAL CONTRIBUTIONS

- **Grant Reviewer:** Served as a travel grant reviewer for the Graduate and Professional Student Association (GPSA) at Arizona State University from May 2022 to August 2023.
- **Conference Volunteer:** Volunteered to organize two conferences - SIGSPATIAL 2022 and SIGMOD 2023.
- **Presentations & Talks:** Five conference presentations - SIGMOD 2023, FOSS4GNA 2023, ICDE 2022, SIGSPATIAL 2022, and ICCIT 2015.
- **Open Source Contribution:** Contributed to Apache Sedona, an open-source geospatial cluster computing framework with 2k+ GitHub Stars, by adding support for two new spatial data types.
- **Membership:** I hold a membership in the professional organizations such as ACM, IEEE, and SIGSPATIAL

PARTICIPATION AND AWARDS

- Recipient of ACM SIGMOD 2023 student travel award to attend the conference and present a paper.
- Recipient of ACM SIGSPATIAL 2022 travel award to attend the conference and present a paper.
- Received ASU Graduate and Professional Student Association (GPSA) travel grant twice - 2022 and 2023
- Recipient of CIDSE Doctoral Fellowship at Arizona State University for the academic year 2018-2019.
- 2nd Runner-up at National Hackathon organized by ICT Division of Bangladesh in 2014. The challenge of the hackathon was to design a project-based solution to solve a national problem of the country.
- 2nd Runner-up at National Mobile Application Code Hub organized by BUET, Bangladesh in 2014.
- Recipient of Honors award from my undergraduate university for maintaining academic excellence.
- Recipient of merit scholarship in all four years of my undergraduate education.
- Recipient of the government merit scholarship in Secondary School Certificate examination, Higher Secondary School Certificate examination, 8th-grade public examination, and 5th-grade public examination.

TEACHING

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| • Instructor
<i>Algorithms (COSC 3100)</i> | Spring 2026
Marquette University |
| • Instructor
<i>Advanced Algorithms (COSC 6260)</i> | Spring 2026
Marquette University |
| • Instructor
<i>Fundamentals of Artificial Intelligence (COSC 4600/5600)</i> | Fall 2025
Marquette University |
| • Instructor
<i>Spatial Data Science and Engineering (CSE 594)</i> | Fall 2022
Arizona State University |
| • Teaching Assistant
<i>Distributed Database Systems (CSE 512)</i> | Fall 2021, Spring 2021, Fall 2020, and Spring 2020
Arizona State University |
| • Teaching Assistant
<i>Data Processing at Scale (CSE 511)</i> | Fall 2019
Arizona State University |
| • Teaching Assistant
<i>Object-Oriented Programming & Data Structure (CSE 205)</i> | Spring 2019
Arizona State University |
| • Teaching Assistant
<i>Principles of Programming with C++ (CSE 100)</i> | Spring 2019
Arizona State University |
| • Teaching Assistant
<i>Principles of Programming with Java & Python (CSE 100)</i> | Fall 2018
Arizona State University |

TECHNICAL SKILLS

- **Programming:** Python, Java, C, C++, Scala, SQL, and HTML
- **Databases:** PostgreSQL, SparkSQL, and MySQL
- **Machine Learning:** PyTorch, Scikit-learn, Keras, TensorFlow, ML & DL Models, and ML Statistics
- **Data Analytics:** Apache Spark, Apache Sedona, PySpark, GeoPandas, Pandas, Matplotlib, and Plotly
- **Teaching:** Course Curriculum and Course Material Development, Managing Courses on Canvas and D2L
- **Others:** Distributed Computing, SDE Design Principles, CI/CD Pipeline, Docker, Jira, and Rest API

PROJECTS

- **GeoTorchAI:** A deep learning and data preprocessing framework for raster imagery and spatiotemporal vector datasets, with **400+ GitHub Stars**. It enables spatiotemporal machine learning practitioners to easily and efficiently implement spatiotemporal deep learning models, besides supporting scalable data preprocessing.
- **InferF:** An out-of-box AI/ML-SQL co-optimization approach for end-to-end inference workflows where the users specify a SQL query and a pre-trained model exported in ONNX format, and the end-to-end processing will be automatically optimized reducing the execution latency.
- **ML-Aware Data Re-partitioning:** This is a framework which aims at reducing the training time and memory usage of a spatial machine learning model by reducing the number of partitions in a spatial grid dataset. Experiments on four datasets achieved a significant reduction in training time and memory usage.
- **NLIDB-Bench:** A benchmark for evaluating state-of-the-art approaches of SQL query generation from natural language queries. It proposes a set of evaluation metrics and conducts experiments with four datasets.